

HDOC Day-39 - String Operations Using Functions and Pointers

1. Algorithm

Step 1: Start the program.

Step 2: Read the student's first name, middle name, and last name.

Step 3: Display the menu with eight choices: combine names, find length, copy full name, compare first and last names, reverse name order, count characters, search a word, and exit.

Step 4: Read the user's choice and use a switch statement.

Step 5: For Choice 1, call **combineNames()**. Copy the first name into fullName using strcpy(), then append spaces, middle name, and last name using strcat().

Step 6: For Choice 2, first form the full name and call **findLength()**. Use strlen() and return the length through a pointer.

Step 7: For Choice 3, first form the full name and call **copyName()**. Copy source into destination using strcpy().

Step 8: For Choice 4, call **compareNames()**. Use strcmp() and return the comparison result through a pointer. Display whether the first and last names are equal or different.

Step 9: For Choice 5, call **reverseNameOrder()**. Create the string in last-name, middle-name, first-name order using strcpy() and strcat().

Step 10: For Choice 6, first form the full name and call **countCharacters()**. Scan each character using strlen(); use isalpha(), isdigit(), isspace(), and tolower() to count vowels, consonants, digits, and spaces. Return all counts through pointers.

Step 11: For Choice 7, read a word and call **searchWord()**. Use strstr() to check whether the word occurs in the full name and return the result through a pointer.

Step 12: For Choice 8, exit the program. For any other choice, display "Invalid choice".

Step 13: Repeat the menu until the user selects Exit.

Step 14: Stop.

2. C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void combineNames(const char firstName[], const char middleName[],
                 const char lastName[], char fullName[])
{
    strcpy(fullName, firstName);
    strcat(fullName, " ");
    strcat(fullName, middleName);
    strcat(fullName, " ");
    strcat(fullName, lastName);
}

void findLength(const char fullName[], size_t *length)
{
    *length = strlen(fullName);
}

void copyName(const char source[], char destination[])
{
    strcpy(destination, source);
}

void compareNames(const char firstName[], const char lastName[],
                  int *comparisonResult)
{

```

```

        *comparisonResult = strcmp(firstName, lastName);
    }

void reverseNameOrder(const char firstName[], const char middleName[],
                     const char lastName[], char reverseName[])
{
    strcpy(reverseName, lastName);
    strcat(reverseName, " ");
    strcat(reverseName, middleName);
    strcat(reverseName, " ");
    strcat(reverseName, firstName);
}

void countCharacters(const char fullName[], int *vowels, int *consonants,
                    int *digits, int *spaces)
{
    int i;
    char ch;
    *vowels = *consonants = *digits = *spaces = 0;

    for (i = 0; i < strlen(fullName); i++) {
        ch = tolower((unsigned char)fullName[i]);
        if (isalpha((unsigned char)ch)) {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u')
                (*vowels)++;
            else
                (*consonants)++;
        }
        else if (isdigit((unsigned char)ch))
            (*digits)++;
        else if (isspace((unsigned char)ch))
            (*spaces)++;
    }
}

void searchWord(const char fullName[], const char word[], int *searchResult)
{
    if (strstr(fullName, word) != NULL)
        *searchResult = 1;
    else
        *searchResult = 0;
}

int main()
{
    char firstName[50], middleName[50], lastName[50];
    char fullName[160], copiedName[160], reverseName[160], word[50];
    int choice, comparisonResult, searchResult;
    int vowels, consonants, digits, spaces;
    size_t length;

    printf("Enter first name: ");
    scanf("%49s", firstName);
    printf("Enter middle name: ");
    scanf("%49s", middleName);
    printf("Enter last name: ");
    scanf("%49s", lastName);

    do {
        printf("\n----- STUDENT NAME OPERATIONS -----\n");
        printf("1. Combine and display full name\n");
        printf("2. Find length of full name\n");
        printf("3. Copy full name\n");
        printf("4. Compare first name and last name\n");
        printf("5. Display name in reverse order\n");
        printf("6. Count vowels, consonants, digits and spaces\n");
        printf("7. Search a word in full name\n");
        printf("8. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                combineNames(firstName, middleName, lastName, fullName);
                printf("Full name: %s\n", fullName);
                break;
            case 2:
                combineNames(firstName, middleName, lastName, fullName);
                findLength(fullName, &length);
                printf("Length of full name: %zu\n", length);
                break;
            case 3:
                combineNames(firstName, middleName, lastName, fullName);
                copyName(fullName, copiedName);
                printf("Copied name: %s\n", copiedName);
                break;
        }
    } while (choice != 8);
}

```

```

        case 4:
            compareNames(firstName, lastName, &comparisonResult);
            if (comparisonResult == 0)
                printf("First name and last name are equal.\n");
            else
                printf("First name and last name are different.\n");
            break;
        case 5:
            reverseNameOrder(firstName, middleName, lastName, reverseName);
            printf("Reverse name order: %s\n", reverseName);
            break;
        case 6:
            combineNames(firstName, middleName, lastName, fullName);
            countCharacters(fullName, &vowels, &consonants, &digits, &spaces);
            printf("Vowels: %d\n", vowels);

            printf("Consonants: %d\n", consonants);
            printf("Digits: %d\n", digits);
            printf("Spaces: %d\n", spaces);
            break;
        case 7:
            combineNames(firstName, middleName, lastName, fullName);
            printf("Enter word to search: ");
            scanf("%49s", word);
            searchWord(fullName, word, &searchResult);
            if (searchResult == 1)
                printf("Word found in full name.\n");
            else
                printf("Word not found in full name.\n");
            break;
        case 8:
            printf("Exiting program...\n");
            break;
        default:
            printf("Invalid choice. Please enter 1 to 8.\n");
    }
} while (choice != 8);
return 0;
}

```

3. Test Case Table

Test Case	Choice	Input / Condition	Expected Output
TC-1	1	Aman Kumar Sharma	Full name: Aman Kumar Sharma
TC-2	2	Aman Kumar Sharma	Length of full name: 17
TC-3	3	Aman Kumar Sharma	Copied name: Aman Kumar Sharma
TC-4	4	Aman Kumar Aman	First and last names are equal
TC-5	4	Aman Kumar Sharma	First and last names are different
TC-6	5	Aman Kumar Sharma	Sharma Kumar Aman
TC-7	6	Aman Kumar Sharma	Vowels: 6, Consonants: 9, Digits: 0, Spaces: 2
TC-8	6	Aman K2 Sharma	Vowels: 5, Consonants: 8, Digits: 1, Spaces: 2
TC-9	7	Search: Kumar	Word found in full name
TC-10	7	Search: Singh	Word not found in full name
TC-11	8	-	Program exits
TC-12	9	-	Invalid choice

4. Test Case Evaluation with Output Snippets

TC-1: combineNames() uses strcpy() and strcat() to join the three names with spaces. The full name is formed correctly. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 1
Full name: Aman Kumar Sharma
```

TC-2: findLength() uses strlen() and stores the result through a pointer. "Aman Kumar Sharma" contains 17 characters including two spaces. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 2
Length of full name: 17
```

TC-3: copyName() copies the complete source string into destination using strcpy(). **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 3
Copied name: Aman Kumar Sharma
```

TC-4: strcmp() returns 0 because the first name and last name are both Aman. The program reports them as equal. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Aman
Enter your choice: 4
First name and last name are equal.
```

TC-5: strcmp() gives a non-zero result for Aman and Sharma, so the program reports them as different. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 4
First name and last name are different.
```

TC-6: reverseNameOrder() joins last name, middle name, and first name in that order. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 5
Reverse name order: Sharma Kumar Aman
```

TC-7: The program scans the full name and correctly counts alphabetic characters and spaces: 6 vowels, 9 consonants, 0 digits, and 2 spaces. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 6
Vowels: 6
Consonants: 9
Digits: 0
Spaces: 2
```

TC-8: The digit 2 is detected by `isdigit()`, while the remaining letters and spaces are counted in their respective categories. **Test Passed.**

```
Enter first name: Aman
Enter middle name: K2
Enter last name: Sharma
Enter your choice: 6
Vowels: 5
Consonants: 8
Digits: 1
Spaces: 2
```

TC-9: `strstr()` finds Kumar inside the full name and the search result is returned through a pointer. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 7
Enter word to search: Kumar
Word found in full name.
```

TC-10: `strstr()` does not find Singh in the full name, so the program displays the not-found message. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 7
Enter word to search: Singh
Word not found in full name.
```

TC-11: Choice 8 executes the exit case and terminates the menu loop. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 8
Exiting program...
```

TC-12: Choice 9 is outside the valid range 1 to 8, so the default case displays an invalid-choice message. **Test Passed.**

```
Enter first name: Aman
Enter middle name: Kumar
Enter last name: Sharma
Enter your choice: 9
Invalid choice. Please enter 1 to 8.
```